THARUN YETTI

Voice Engineer · LiveKit / SIP Telephony / Real-Time Voice AI · Compliance-Driven Platforms
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PROFESSIONAL SUMMARY

Backend engineer with 1+ year building real-time voice AI systems in regulated industries. Own SIP/PSTN telephony (LiveKit, Twilio, Vobiz), multi-provider STT/TTS orchestration, and call routing/escalation logic in a live healthcare platform, backed by a full observability stack (Grafana/Prometheus/OpenTelemetry/Langfuse). Looking for a Voice Engineer role owning the voice stack end-to-end.

EXPERIENCE

Member of Technical Staff - *AmbitOS (AI Prof Private Ltd)* May 2026 - Present

- Own SIP/PSTN telephony integration end-to-end on LiveKit, running Twilio and Vobiz in production — handling 500+ patient voice calls with 50–80 concurrent call capacity and a 70% pickup rate.
- Built and tuned STT/TTS pipelines across multiple providers (Cartesia, Soniox, OpenAI) and evaluated speech-to-speech realtime models to cut pipeline latency and provider lock-in.
- Handle call routing and dispatch logic — queue management, load balancing across LiveKit agent instances, and failover/forwarding to available agents.
- Built human-escalation workflow where the AI agent raises a handoff request and routes the call to an available human when the patient asks and then the authorized person can join the call or else call later.
- Built a full observability stack on AWS — Grafana, Prometheus, Loki, OpenTelemetry, Langfuse, CloudWatch — maintaining <3% error rate across the voice pipeline.
- Built and maintained CI/CD pipelines (GitHub Actions) with automated security scanning gating deployments from staging to production.
- Build and maintain secure backend APIs in FastAPI with cryptographic integrations - encryption at rest/in transit and key management.

Full Stack Developer Intern - *Hitloop (now AmbitOS)* May 2025 - Apr 2026

- Contributed to a healthcare system engineered for compliance, covering PHI access control, audit logging, and data residency.
- Built role-based dashboard features and integrated real-time voice AI into a patient-facing conversational agent.
- Built secure, scalable backend APIs in FastAPI, including cryptographic integrations.
- Cut real-time interaction latency ~30% by redesigning event handling and API flow.

PROJECTS

Resume-Hub | Live Demo - *MERN, Gemini API*

- Built a full-stack resume analysis platform providing ATS compatibility checks, job recommendations, and instant interview-question generation.
- Integrated the Gemini API for resume parsing and AI-driven insights.

Back-End Application for Shopping Cart | GitHub - *Spring Boot Microservices, Docker, Kubernetes*

- Designed a microservices-based backend with RESTful APIs for core shopping-cart functionality.
- Containerized services with Docker and orchestrated deployments with Kubernetes, focused on scalability and fault isolation.

University Discussion & Collaboration Platform | GitHub - *MERN, Tailwind CSS, Material UI*

- Built a responsive platform for student-mentor discussions, scheduling, and result publication.
- Secured 1st place in a college-level hackathon.

Heart Disease Classification Using ML | GitHub - *Scikit-learn, Pandas, NumPy*

- Built an ML pipeline predicting heart disease from clinical parameters, using a majority-voting ensemble across multiple classifiers to improve reliability.

SKILLS

Languages: Java, C, Python, JavaScript

Telephony & Real-Time Voice: LiveKit, WebRTC, SIP Trunking, Twilio, Vobiz, Call Routing, Barge-in/Turn Detection

STT/TTS: Cartesia, Soniox, OpenAI Realtime, Deepgram, Speech-to-Speech models

Backend & Frameworks: FastAPI, Node.js, REST APIs, Spring Boot (Microservices)

Frontend: React.js, Tailwind CSS, Material UI

DevOps & Cloud: Docker, Kubernetes, CI/CD, Microservices Architecture, Nginx, AWS

Databases & Caching: MySQL, PostgreSQL, MongoDB, Redis

Compliance & Security: HIPAA, GDPR, SOC2, ISO 27001, RBAC, Encryption (At-Rest/In-Transit), Audit Logging

Observability: Grafana, Prometheus, Loki, Promtail, OpenTelemetry, Langfuse, Alerting, CloudWatch

Tools & Platforms: Git, GitHub, IntelliJ, VSCode, Jupyter Notebook

EDUCATION

B.Tech in Computer Science, CGPA: 9.43 - *RGUKT, IIIT-RK Valley, AP, India* 2022 - 2026

Pre-University Course (Equivalent to Intermediate), CGPA: 9.94 - *RGUKT, IIIT-RK Valley, AP, India* 2020 - 2022

ACHIEVEMENTS

- 1st and 3rd place in campus hackathons (2025 & 2024).
- AIR 6009 in GATE CS 2025 with 519 score.
- 450+ DSA problems solved on LeetCode.